

Yucel Yavuz

Director, AI & Data Science | Generative AI | Agentic AI | AI Product Leadership

New Jersey, USA | yyavuz.ds@gmail.com | (732) 595-1777 | [linkedin.com/in/yucyavuz](https://www.linkedin.com/in/yucyavuz) | github.com/yucel-yavuz

SUMMARY

Director-level AI and Data Science leader with 7+ years of hands-on experience in machine learning, Generative AI, Agentic AI, and AI product development, backed by 10+ years of cross-functional leadership. Lead and mentor senior data scientists while partnering with executive and cross-functional stakeholders to translate business needs into scalable AI products, decision intelligence, and workflow automation. Experienced in architecting and deploying end-to-end AI solutions using Python, Databricks, Snowflake, Claude Code, and modern LLM frameworks, including AI agents, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), predictive modeling, explainable AI (SHAP), and production ML pipelines.

CORE EXPERTISE

AI Leadership & Strategy: AI strategy • AI product development • Generative AI • Agentic AI • Solution architecture • Decision intelligence

AI Engineering & Platforms: Databricks • Snowflake • Python • SQL • LangChain • LangGraph • RAG • MLflow • Streamlit • Docker • CI/CD

Machine Learning & Analytics: Predictive modeling • Explainable AI (SHAP) • Segmentation • Causal inference • A/B testing • Time Series Forecasting

Business Intelligence: Tableau • Power BI • Plotly

Leadership: People leadership • Cross-functional leadership • Executive communication • Stakeholder management

Representative Business Impact: 35% faster stakeholder decision-making • 40% reduction in manual reporting

• 92% predictive model accuracy • 15% improvement in campaign efficiency • 12% increase in customer retention

• \$150K operational savings

PROFESSIONAL EXPERIENCE

Horizon Media – New York, NY

Director, AI & Data Science | Generative AI | Apr 2023 – Present

- Lead and mentor senior data scientists while partnering with technical and non-technical stakeholders to deliver ML and GenAI solutions from concept through production.
- Accelerated stakeholder decision-making by 35% and reduced manual reporting by 40% by architecting LLM-driven analytics solutions with LangChain, LangGraph, Streamlit, and Snowflake Cortex.
- Increased model transparency and stakeholder adoption by 20% across analytics workflows by leading explainable AI frameworks on Databricks using SHAP.
- Improved campaign efficiency by 15% and achieved 92% model accuracy by designing and operationalizing Databricks-Snowflake ML pipelines for LTV and look-alike modeling using XGBoost and ensemble methods.
- Improved customer lifecycle performance by approximately 10% by building scalable audience intelligence frameworks integrating behavioral, transactional, and temporal data to uncover conversion drivers.
- Improved targeting performance by approximately 12% by applying causal inference methodologies (A/B testing, propensity score matching) to quantify campaign lift and ROI and strengthen media investment decisions.
- Contributed to two major client wins by delivering ML- and GenAI-driven audience profiling frameworks that strengthened audience strategy and new business pitches.
- Increased retention by 12% by directing segmentation and behavioral analytics initiatives in Databricks to identify high-value customer segments.
- Reduced insight turnaround by 30% and enabled real-time strategic planning by designing interactive Tableau and Power BI dashboards for 1P metrics, cross-sell trends, and behavioral clusters.

Bowery Farming – Kearny, NJ

Data Scientist | Operations Manager | Sep 2018 – Apr 2023

- Improved yield accuracy by 10% and strengthened production planning by developing predictive yield models across 7 ML and 1 DL algorithm.
- Enhanced productivity by 25% and warehouse efficiency by 20% by building automated production dashboards.
- Reduced packaging losses by 20% by introducing machine learning-driven process optimization.
- Promoted 6 employees and reduced turnover to 2% by creating a Team Lead Development Tool and training program.
- Maintained production compliance with company quality, sanitation, and food safety standards by fostering a team culture that supported SQF results of 98% in 2019 and 2020 and 99% in 2021.

Teleplan, Mechanicsburg, PA

Sr. Analyst | Supply Chain | Jan 2017 – Sep 2018

- Improved efficiency by \$150K in 8 months and increased supplier performance to 92.6% by developing a supply chain analytics dashboard.
- Achieved 99.8% cycle count accuracy by streamlining forecasting, procurement, and vendor workflows.
- Optimized material reuse and cost savings initiatives by introducing analytical process improvements.

Turkish Army, Ankara/Turkey

Operations and Analytics Manager | Aug 2001-April 2016

- Saved \$300K in resource allocation by spearheading 10+ strategic analytics projects.
- Supported executive decision-making across multiple departments by designing regression-based forecasting models for personnel and logistics planning and by delivering reports and dashboards.

RECENT PROJECTS

- **AI Media Planning Agent – Multi-Agent LLM System for Marketing Strategy**

Python, LangGraph, LangChain, OpenAI, Streamlit

- Automated end-to-end marketing workflows by architecting a multi-agent AI platform that turns natural language prompts into business insights, audience strategy, media plans, creative recommendations, and personalized communications.

- **Business Intelligence Chatbot – Text-to-SQL AI Platform (Databricks)**

Python, Streamlit, Databricks AI (ChatDatabricks), LangChain, LangGraph, Plotly

- Enabled self-service analytics by designing and deploying a conversational BI platform on Databricks that translates natural language into SQL, executes analytics workflows, and delivers interactive visual insights.

- **SHAP Analysis Workflow – End-to-End Explainable AI Product on Databricks**

Python, XGBoost, SHAP, Streamlit, Databricks AI, Databricks Feature Store, Snowflake

- Delivered scalable stakeholder decision intelligence by architecting an explainable AI product that combines XGBoost modeling, SHAP driver analysis, and GenAI-powered executive reporting.

- **Affinity Engine – AI-Powered Product Affinity Platform**

Python, Streamlit, Databricks AI, Snowflake, Plotly

- Architected an AI product that analyzes large-scale customer transaction and audience data to identify product affinity, cross-sell opportunities, and audience expansion strategies for media and growth planning.

- **Cluster Analysis Workflow – Audience Segmentation (GenAI) Powered on Databricks**

Python, PySpark MLlib, Databricks AI, Snowflake

- Designed an automated audience segmentation pipeline leveraging PCA, clustering, and GenAI-powered executive reporting to generate customer personas, behavioral insights, and activation strategies.

- **Marketing Assistant CoPilot – RAG-Based Knowledge Assistant**

Python, Streamlit, LangChain, OpenAI, FAISS

- Enabled natural-language document querying and summarization by building a RAG-powered knowledge assistant for enterprise documents.

EDUCATION

- **U.S. Army War College – Carlisle, PA**

International Fellows Academic Prep Course

- **Turkish National Defense University – Ankara, Turkey**

Ph.D. (Pending), Technology Management

M.S., Technology Management

B.S., Systems Engineering

- **Turkish Army War College – Istanbul, Turkey**

M.A., Organizational Management & Leadership

CERTIFICATIONS

- Generative AI for Data Scientists – *IBM*
- Generative AI Bootcamp – *Business Science*
- Master Machine Learning for Data Science & AI - *CodeBasics*
- Databricks Mosaic AI – *Coursera*
- Certified Management Executive (CME) – *Management & Strategy Institute*